

Python Conditionals and Operations

100 Progressive Programming Problems

Problems 1-25	Basic arithmetic and simple if/else
Problems 26-50	if/elif/else and multiple ranges
Problems 51-75	Combined conditions, logical operators, and nested decisions
Problems 76-100	Multi-stage calculations and realistic decision systems

Classroom worksheet compilation • Problems only • No code solutions

SECTION I

Problems 1-25: Foundations

Basic arithmetic, input/output, comparisons, and simple if/else decisions.

Problem 1: Electricity Bill Calculator

Objective: Create a Python program that calculates an electricity bill based on the number of kilowatt-hours consumed.

Input:

Enter kWh consumed:

Rates:

- 0 or below → ₱0
- 1-100 kWh → ₱10 per kWh
- 101-200 kWh → ₱12 per kWh
- Above 200 kWh → ₱15 per kWh

After calculating the bill, ask:

Are you a senior citizen? (y/n):

- y → subtract ₱100
- n → no discount
- Final bill cannot be below ₱0.

Display:

Bill before discount: ₱__

Final Bill: ₱__

Test Cases:

kWh	Senior?	Expected Final Bill
0	n	₱0
80	n	₱800
150	n	₱1,800
250	n	₱3,750
150	y	₱1,700

Problem 2: Water Bill Calculator

Objective: Calculate a household's water bill.

Input:

Enter cubic meters consumed:

Rates:

- 0 or below → ₱0
- 1-20 m³ → ₱25 per m³

- Above 20 m³ → ₱35 per m³

Ask:

Is this a residential account? (y/n):

- y → ₱50 discount
- n → no discount

Example Test Cases:

- 10 m³, n → ₱250
- 20 m³, y → ₱450
- 30 m³, n → ₱1,050

Problem 3: Student Passing Checker

Objective: Determine whether a student passed or failed.

Input:

Enter final grade:

Conditions:

- Grade 75 or higher → Passed
- Grade below 75 → Failed

Also calculate how many points the student is above or below the passing grade.

Examples:

- 85 → Passed by 10 points
- 75 → Passed by 0 points
- 68 → Failed by 7 points

Problem 4: Shopping Discount Calculator

Objective: Calculate the final price of a purchase.

Input:

Enter total purchase amount:

Conditions:

- ₱1,000 or more → 10% discount
- Below ₱1,000 → no discount

Display:

- Original price
- Discount
- Final price

Examples:

- ₱800 → ₱800
- ₱1,000 → ₱900

- ₱2,500 → ₱2,250

Problem 5: Employee Overtime Pay

Objective: Calculate an employee's weekly salary.

Inputs:

- Hours worked
- Hourly rate

Conditions:

- 40 hours or less → normal pay
- More than 40 hours → hours above 40 are paid at 1.5× hourly rate

Examples:

- 40 hours × ₱100 → ₱4,000
- 45 hours × ₱100 → ₱4,750
- 50 hours × ₱80 → ₱4,600

Problem 6: Movie Ticket Price

Objective: Determine ticket price based on age.

Input:

Enter age:

Prices:

- Age 0-12 → ₱150
- Age 13 and above → ₱250

Ask:

Is it a weekday? (y/n):

- **y** → ₱20 discount
- **n** → regular price

Examples:

- Age 10, weekend → ₱150
- Age 10, weekday → ₱130
- Age 20, weekday → ₱230

Problem 7: Temperature Checker

Objective: Classify a temperature.

Input:

Enter temperature in Celsius:

Conditions:

- Below 20°C → **Cold**

- 20°C and above → Warm

Also convert Celsius to Fahrenheit using:

$$F = (C \times 9 / 5) + 32$$

Examples:

- 10°C → Cold, 50°F
- 20°C → Warm, 68°F
- 30°C → Warm, 86°F

Problem 8: Delivery Fee Calculator

Objective: Calculate a delivery charge based on distance.

Input:

Enter delivery distance in km:

Rates:

- 5 km or less → ₱50
- More than 5 km → ₱10 per kilometer

Ask whether the customer has free-delivery membership.

- y → delivery fee = ₱0
- n → normal fee

Examples:

- 4 km, n → ₱50
- 8 km, n → ₱80
- 15 km, y → ₱0

Problem 9: Number Sign Checker

Objective: Determine whether a number is positive, negative, or zero.

Input:

Enter a number:

Conditions:

- Greater than 0 → Positive
- Less than 0 → Negative
- Equal to 0 → Zero

Also display the square of the number.

Examples:

- 5 → Positive, square = 25
- -4 → Negative, square = 16
- 0 → Zero, square = 0

Problem 10: Even or Odd Checker

Objective: Determine whether an integer is even or odd.

Input:

Enter an integer:

Use the modulus operator %.

Conditions:

- $\text{Number} \% 2 == 0 \rightarrow \text{Even}$
- Otherwise $\rightarrow \text{Odd}$

Also display twice the entered number.

Examples:

- 10 $\rightarrow$ Even, doubled = 20
- 7 $\rightarrow$ Odd, doubled = 14
- -4 $\rightarrow$ Even, doubled = -8

Problem 11: Restaurant Service Charge

Objective: Calculate a restaurant bill.

Input:

Enter food bill:

Conditions:

- Bill below ₱1,000 $\rightarrow$ no service charge
- Bill ₱1,000 or above $\rightarrow$ add 10% service charge

Display:

- Food bill
- Service charge
- Total bill

Examples:

- ₱800 $\rightarrow$ ₱800
- ₱1,000 $\rightarrow$ ₱1,100
- ₱2,500 $\rightarrow$ ₱2,750

Problem 12: Free Shipping Checker

Objective: Determine a customer's final online shopping cost.

Input:

Enter order amount:

Conditions:

- ₱1,500 or higher $\rightarrow$ free shipping
- Below ₱1,500 $\rightarrow$ add ₱120 shipping fee

Examples:

- ₱500 → ₱620
- ₱1,499 → ₱1,619
- ₱1,500 → ₱1,500

Problem 13: Salary Bonus Calculator

Objective: Calculate an employee's bonus.

Inputs:

- Monthly salary
- Years of service

Conditions:

- 5 years or more → bonus = 10% of salary
- Less than 5 years → bonus = 5% of salary

Display:

Salary, bonus, and total compensation.

Examples:

- ₱20,000, 2 years → bonus ₱1,000
- ₱20,000, 5 years → bonus ₱2,000
- ₱30,000, 10 years → bonus ₱3,000

Problem 14: Parking Fee Calculator

Objective: Calculate parking charges.

Input:

Enter number of hours parked:

Rates:

- 2 hours or less → ₱50
- More than 2 hours → ₱50 + ₱20 for every additional hour

Examples:

- 1 hour → ₱50
- 2 hours → ₱50
- 5 hours → ₱110

Problem 15: Exam Eligibility Checker

Objective: Determine whether a student may take an examination.

Inputs:

- Number of classes held
- Number of classes attended

Calculate:

$$\text{Attendance Percentage} = \text{attended} / \text{held} \times 100$$

Condition:

- 75% or higher → **Eligible**
- Below 75% → **Not Eligible**

Examples:

- 100 held, 80 attended → 80%, Eligible
- 100 held, 74 attended → 74%, Not Eligible
- 40 held, 30 attended → 75%, Eligible

Problem 16: Mobile Data Charge

Objective: Calculate a customer's mobile-data bill.

Input:

Enter GB used:

Rates:

- 5 GB or less → ₱300
- More than 5 GB → ₱300 + ₱50 per additional GB

Examples:

- 3 GB → ₱300
- 5 GB → ₱300
- 8 GB → ₱450

Problem 17: Bank Withdrawal Checker

Objective: Determine whether a withdrawal can be completed.

Inputs:

- Current balance
- Withdrawal amount

Conditions:

- If withdrawal $\leq$ balance → approve
- If withdrawal $>$ balance → display **Insufficient Balance**

If approved:

$$\text{New Balance} = \text{Current Balance} - \text{Withdrawal}$$

Examples:

- Balance ₱5,000, withdraw ₱2,000 → ₱3,000
- Balance ₱1,000, withdraw ₱1,000 → ₱0
- Balance ₱1,000, withdraw ₱1,500 → Insufficient Balance

Problem 18: Sales Commission Calculator

Objective: Calculate a salesperson's commission.

Input:

Enter total sales:

Conditions:

- Sales below ₱10,000 → 5% commission
- Sales ₱10,000 or more → 10% commission

Display:

- Total sales
- Commission rate
- Commission earned

Examples:

- ₱5,000 → ₱250
- ₱10,000 → ₱1,000
- ₱25,000 → ₱2,500

Problem 19: Simple Income Tax

Objective: Calculate income tax.

Input:

Enter monthly income:

Conditions:

- ₱20,000 or less → no tax
- More than ₱20,000 → 10% tax

Calculate:

Net Income = Income - Tax

Examples:

- ₱15,000 → tax ₱0
- ₱20,000 → tax ₱0
- ₱30,000 → tax ₱3,000; net ₱27,000

Problem 20: Product Stock Checker

Objective: Determine whether an order can be completed.

Inputs:

- Available stock
- Quantity ordered
- Price per item

Conditions:

- Quantity ordered $\leq$ available stock $\rightarrow$ process order
- Quantity ordered $>$ available stock $\rightarrow$ Not enough stock

If processed:

$$\text{Total Cost} = \text{Quantity} \times \text{Price}$$

Examples:

- Stock 20, order 5, ₱100 $\rightarrow$ ₱500
- Stock 10, order 10, ₱50 $\rightarrow$ ₱500
- Stock 5, order 8 $\rightarrow$ Not enough stock

Problem 21: BMI Classification

Objective: Calculate Body Mass Index and give a simple classification.

Inputs:

- Weight in kilograms
- Height in meters

Formula:

$$\text{BMI} = \text{weight} / \text{height}^2$$

For this introductory exercise:

- BMI below 18.5 $\rightarrow$ Below Normal Range
- BMI 18.5 or higher $\rightarrow$ Normal or Above

Examples:

- 50 kg, 1.70 m $\rightarrow$ BMI $\approx$ 17.30 $\rightarrow$ Below Normal Range
- 60 kg, 1.70 m $\rightarrow$ BMI $\approx$ 20.76 $\rightarrow$ Normal or Above
- 80 kg, 1.80 m $\rightarrow$ BMI $\approx$ 24.69 $\rightarrow$ Normal or Above

Problem 22: Loan Qualification Checker

Objective: Determine whether an applicant qualifies for a basic loan.

Inputs:

- Monthly salary
- Desired loan amount

Condition:

A person's maximum allowed loan is:

$$\text{Maximum Loan} = \text{Monthly Salary} \times 10$$

- Requested amount $\leq$ maximum loan $\rightarrow$ Loan Qualified
- Requested amount $>$ maximum loan $\rightarrow$ Loan Not Qualified

Examples:

- Salary ₱20,000, loan ₱100,000 $\rightarrow$ Qualified
- Salary ₱20,000, loan ₱200,000 $\rightarrow$ Qualified

- Salary ₱20,000, loan ₱250,000 → Not Qualified

Problem 23: Bus Fare Calculator

Objective: Calculate a passenger's bus fare.

Inputs:

- Distance traveled in kilometers
- Student status (y/n)

Rate:

Base Fare = Distance × ₱3

If the passenger is a student:

- Apply a 20% discount

Otherwise:

- Charge full fare

Examples:

- 10 km, n → ₱30
- 10 km, y → ₱24
- 50 km, y → ₱120

Problem 24: Store Membership Discount

Objective: Calculate a customer's final purchase price.

Inputs:

- Purchase amount
- Membership status (y/n)

Conditions:

- Member → 15% discount
- Non-member → no discount

Calculate:

Discount = Purchase × Discount Rate

Final Price = Purchase - Discount

Examples:

- ₱1,000, n → ₱1,000
- ₱1,000, y → ₱850
- ₱5,000, y → ₱4,250

Problem 25: Simple Grade Calculator

Objective: Calculate a student's average and determine whether the student passed.

Inputs:

- Quiz score
- Project score
- Exam score

Calculate:

$$\text{Average} = (\text{Quiz} + \text{Project} + \text{Exam}) / 3$$

Conditions:

- Average $\geq 75 \rightarrow$ PASSED
- Average $< 75 \rightarrow$ FAILED

Display:

- Quiz score
- Project score
- Exam score
- Average
- Result

Test Cases:

Quiz	Project	Exam	Average	Result
80	85	90	85.00	PASSED
75	75	75	75.00	PASSED
60	70	80	70.00	FAILED
90	80	70	80.00	PASSED
50	60	70	60.00	FAILED

These first 25 intentionally keep the logic fairly simple. A good progression is for Problems 26–50 to introduce more `if/elif/else` ranges, such as multiple tax brackets, tiered fares, letter grades, hotel pricing, calorie classifications, ATM fees, phone plans, and multi-level discounts.

SECTION II

Problems 26-50: Multiple Ranges

if/elif/else structures, tiered ranges, and progressively richer calculations.

Problem 26: Letter Grade Calculator

Objective: Create a Python program that determines a student's letter grade.

Input:

Enter final grade:

Conditions:

- 90–100 → A
- 85–89 → B
- 80–84 → C
- 75–79 → D
- Below 75 → F

Display:

- Numerical grade
- Letter grade
- Passed if grade is 75 or above
- Failed otherwise

Test Cases:

Grade	Letter	Result
95	A	Passed
87	B	Passed
82	C	Passed
76	D	Passed
70	F	Failed

Problem 27: Tiered Water Consumption Bill

Objective: Calculate a water bill based on consumption.

Input:

Enter water consumption in cubic meters:

Rates:

- 0 or below → ₱0
- 1–15 m³ → ₱20 per m³
- 16–30 m³ → ₱25 per m³
- Above 30 m³ → ₱30 per m³

Display:

- Consumption
- Rate used
- Total water bill

Test Cases:

- 10 m³ → ₱200
- 20 m³ → ₱500
- 30 m³ → ₱750
- 40 m³ → ₱1,200

Problem 28: Internet Plan Overage Calculator

Objective: Calculate a customer's internet bill based on data usage.

Input:

Enter GB used:

Charges:

- Up to 100 GB → ₱999
- 101-200 GB → ₱1,299
- Above 200 GB → ₱1,299 plus ₱10 for every GB above 200

Examples:

- 80 GB → ₱999
- 150 GB → ₱1,299
- 200 GB → ₱1,299
- 250 GB → ₱1,799

Problem 29: Employee Salary Level

Objective: Classify an employee based on monthly salary.

Input:

Enter monthly salary:

Conditions:

- Below ₱15,000 → [Entry Level](#)
- ₱15,000-₱29,999 → [Junior Level](#)
- ₱30,000-₱49,999 → [Mid Level](#)
- ₱50,000 and above → [Senior Level](#)

Also calculate the annual salary:

Annual Salary = Monthly Salary × 12

Problem 30: Parking Fee by Duration

Objective: Calculate parking fees using multiple time ranges.

Input:

Enter number of hours parked:

Rates:

- 1-2 hours → ₱50
- 3-5 hours → ₱100
- 6-10 hours → ₱150
- More than 10 hours → ₱250

Examples:

- 2 hours → ₱50
- 4 hours → ₱100
- 8 hours → ₱150
- 12 hours → ₱250

Problem 31: Restaurant Discount by Bill Amount

Objective: Calculate a discount based on the customer's total food bill.

Input:

Enter total bill:

Discounts:

- Below ₱1,000 → 0%
- ₱1,000-₱2,499 → 5%
- ₱2,500-₱4,999 → 10%
- ₱5,000 and above → 15%

Calculate:

Discount = Bill × Discount Rate

Final Bill = Bill - Discount

Problem 32: Age Group Classifier

Objective: Determine a person's age group.

Input:

Enter age:

Conditions:

- 0-12 → Child
- 13-19 → Teenager
- 20-39 → Young Adult
- 40-59 → Adult

- 60 and above → Senior Citizen

If the age is negative, display:

Invalid Age

Problem 33: Cinema Ticket Calculator

Objective: Calculate a movie ticket price based on age.

Input:

Enter age:

Ticket Prices:

- 0-12 → ₱150
- 13-59 → ₱250
- 60 and above → ₱180

Ask:

Is this a 3D movie? (y/n):

- y → add ₱50
- n → no additional charge

Examples:

- Age 10, n → ₱150
- Age 20, y → ₱300
- Age 65, y → ₱230

Problem 34: Shipping Fee by Weight

Objective: Calculate the shipping fee for a package.

Input:

Enter package weight in kg:

Rates:

- Up to 1 kg → ₱80
- More than 1 kg up to 5 kg → ₱150
- More than 5 kg up to 10 kg → ₱250
- Above 10 kg → ₱400

Display:

Weight and shipping fee.

Problem 35: Exam Performance Rating

Objective: Determine a student's performance based on exam score.

Input:

Enter exam score:

Conditions:

- 90-100 → Excellent
- 80-89 → Very Good
- 75-79 → Good
- 60-74 → Needs Improvement
- Below 60 → Poor

Also display whether the student passed using 75 as the passing score.

Problem 36: Electricity Usage Classification

Objective: Classify electricity consumption and calculate the corresponding bill.

Input:

Enter kWh consumed:

Rates:

- 1-100 kWh → ₱9 per kWh
- 101-200 kWh → ₱11 per kWh
- 201-300 kWh → ₱13 per kWh
- Above 300 kWh → ₱15 per kWh

Classification:

- 1-100 → Low Usage
- 101-200 → Moderate Usage
- 201-300 → High Usage
- Above 300 → Very High Usage

Problem 37: Sales Commission by Performance

Objective: Calculate sales commission using multiple sales ranges.

Input:

Enter monthly sales:

Commission Rates:

- Below ₱10,000 → 3%
- ₱10,000-₱29,999 → 5%
- ₱30,000-₱49,999 → 8%
- ₱50,000 and above → 12%

Display:

- Sales
- Commission rate
- Commission amount

Problem 38: Mobile Load Bonus Calculator

Objective: Calculate the bonus load received by a customer.

Input:

Enter load amount:

Bonuses:

- Below ₱100 → no bonus
- ₱100–₱299 → 5% bonus
- ₱300–₱499 → 10% bonus
- ₱500 and above → 15% bonus

Calculate:

Bonus = Load × Bonus Rate

Total Load = Load + Bonus

Problem 39: Hotel Room Rate Calculator

Objective: Determine the total hotel cost.

Inputs:

- Number of nights
- Room type

Room Types:

- 1 → Standard = ₱1,500/night
- 2 → Deluxe = ₱2,500/night
- 3 → Suite = ₱4,000/night

Condition:

- Stay 7 nights or more → 10% discount

Display:

- Room rate
- Original total
- Discount
- Final total

Problem 40: Loan Interest Calculator

Objective: Calculate simple loan interest based on the amount borrowed.

Input:

Enter loan amount:

Interest Rates:

- Up to ₱10,000 → 5%
- ₱10,001–₱50,000 → 8%

- ₱50,001–₱100,000 → 10%
- Above ₱100,000 → 12%

Calculate:

Interest = Loan × Rate

Total Payment = Loan + Interest

Problem 41: Temperature Classification

Objective: Classify temperature into different weather conditions.

Input:

Enter temperature in Celsius:

Conditions:

- Below 10°C → [Very Cold](#)
- 10–19°C → [Cold](#)
- 20–29°C → [Warm](#)
- 30–39°C → [Hot](#)
- 40°C and above → [Very Hot](#)

Also convert the temperature to Fahrenheit.

Problem 42: Grocery Discount Calculator

Objective: Calculate grocery discounts based on purchase amount.

Input:

Enter grocery total:

Discounts:

- Below ₱500 → 0%
- ₱500–₱999 → 3%
- ₱1,000–₱1,999 → 5%
- ₱2,000 and above → 10%

Ask:

Do you have a loyalty card? (y/n):

If [y](#), apply an additional ₱50 discount after the percentage discount.

The final amount must not fall below ₱0.

Problem 43: Taxi Fare Calculator

Objective: Calculate taxi fare based on distance traveled.

Input:

Enter distance traveled in km:

Rates:

- Up to 3 km → ₱50
- 4-10 km → ₱50 + ₱12 per km beyond 3
- Above 10 km → ₱134 + ₱15 per km beyond 10

Examples:

- 3 km → ₱50
- 5 km → ₱74
- 10 km → ₱134
- 15 km → ₱209

Problem 44: Employee Performance Bonus

Objective: Calculate an employee's bonus based on performance rating.

Inputs:

- Monthly salary
- Performance rating from 1 to 5

Bonus Rates:

- Rating 1 → 0%
- Rating 2 → 3%
- Rating 3 → 5%
- Rating 4 → 8%
- Rating 5 → 10%

If the rating is outside 1-5, display:

Invalid Rating

Calculate:

Bonus = Salary × Bonus Rate

Problem 45: ATM Withdrawal Fee

Objective: Calculate ATM transaction fees.

Input:

Enter withdrawal amount:

Fees:

- Up to ₱1,000 → ₱10
- ₱1,001-₱5,000 → ₱15
- ₱5,001-₱10,000 → ₱20
- Above ₱10,000 → ₱30

Display:

Total Deduction = Withdrawal + Fee

Ask for the user's balance and ensure the account has enough money for the complete deduction.

Problem 46: Scholarship Qualification

Objective: Determine a student's scholarship level.

Inputs:

- General average
- Family monthly income

Conditions:

- Average ≥ 95 and income $\leq \text{P}20,000 \rightarrow$ Full Scholarship
- Average ≥ 90 and income $\leq \text{P}30,000 \rightarrow$ 75% Scholarship
- Average ≥ 85 and income $\leq \text{P}40,000 \rightarrow$ 50% Scholarship
- Otherwise $\rightarrow$ No Scholarship

Test Cases:

Average	Income	Result
97	P18,000	Full Scholarship
92	P25,000	75% Scholarship
87	P35,000	50% Scholarship
80	P20,000	No Scholarship

Problem 47: Phone Plan Calculator

Objective: Determine a customer's monthly phone bill.

Input:

Enter minutes used:

Plans:

- 0-100 minutes $\rightarrow$ P300
- 101-300 minutes $\rightarrow$ P500
- 301-500 minutes $\rightarrow$ P700
- Above 500 minutes $\rightarrow$ P700 + P2 for each minute above 500

Examples:

- 80 minutes $\rightarrow$ P300
- 200 minutes $\rightarrow$ P500
- 450 minutes $\rightarrow$ P700
- 600 minutes $\rightarrow$ P900

Problem 48: Delivery Service Calculator

Objective: Calculate a delivery fee based on both distance and package weight.

Inputs:

- Distance in kilometers
- Package weight in kilograms

Distance Fee:

- Up to 5 km → ₱50
- 6–15 km → ₱100
- More than 15 km → ₱150

Weight Charge:

- Up to 2 kg → ₱0
- More than 2 kg up to 5 kg → add ₱50
- Above 5 kg → add ₱100

Calculate:

Final Delivery Fee = Distance Fee + Weight Charge

Examples:

- 4 km, 1 kg → ₱50
- 10 km, 4 kg → ₱150
- 20 km, 8 kg → ₱250

Problem 49: Electricity Bill With Surcharge

Objective: Calculate an electricity bill and apply a surcharge when necessary.

Input:

Enter kWh consumed:

Rates:

- 1–100 kWh → ₱8 per kWh
- 101–200 kWh → ₱10 per kWh
- 201–300 kWh → ₱12 per kWh
- Above 300 kWh → ₱14 per kWh

After calculating the bill:

- Bill below ₱2,000 → no surcharge
- ₱2,000–₱3,999 → add 5%
- ₱4,000 and above → add 10%

Display:

- Basic bill
- Surcharge
- Final bill

Problem 50: Employee Payroll Calculator

Objective: Create a payroll calculator using salary, overtime, and deductions.

Inputs:

- Basic monthly salary
- Overtime hours

Overtime Rates:

- 0 overtime hours → ₱0
- 1-10 hours → ₱100 per hour
- 11-20 hours → ₱125 per hour
- More than 20 hours → ₱150 per hour

Calculate:

$$\text{Gross Pay} = \text{Basic Salary} + \text{Overtime Pay}$$

Deduction:

- Gross pay below ₱20,000 → 5%
- ₱20,000–₱39,999 → 8%
- ₱40,000 and above → 10%

Then calculate:

$$\text{Deduction} = \text{Gross Pay} \times \text{Deduction Rate}$$
$$\text{Net Pay} = \text{Gross Pay} - \text{Deduction}$$

Test Cases:

Salary	OT Hours	OT Pay	Deduction Rate
₱15,000	0	₱0	5%
₱18,000	5	₱500	5%
₱25,000	10	₱1,000	8%
₱35,000	15	₱1,875	8%
₱45,000	25	₱3,750	10%

For the next set, Problems 51–75 should increase the difficulty noticeably by combining two or more conditions with **and**, **or**, nested **if** statements, validation, multiple discounts/surcharges, and calculations that depend on earlier decisions.

SECTION III

Problems 51-75: Combined Logic

Logical operators, nested decisions, validation, and interacting conditions.

Problem 51: Student Honors Qualification

Objective: Determine whether a student qualifies for academic honors.

Inputs:

- General average
- Conduct grade
- Number of absences

Conditions:

- Average ≥ 95 , conduct ≥ 90 , and absences $\leq 3 \rightarrow$ With Highest Honors
- Average ≥ 90 , conduct ≥ 85 , and absences $\leq 5 \rightarrow$ With High Honors
- Average ≥ 85 , conduct ≥ 80 , and absences $\leq 7 \rightarrow$ With Honors
- Otherwise $\rightarrow$ No Honors

Test Cases:

Average	Conduct	Absences	Result
97	95	2	With Highest Honors
92	90	4	With High Honors
87	85	6	With Honors
95	75	2	No Honors

Problem 52: Employee Bonus With Attendance

Objective: Calculate an employee's bonus using performance and attendance.

Inputs:

- Monthly salary
- Performance rating from 1-5
- Number of absences

Bonus Rates:

- Rating 5 $\rightarrow$ 15%
- Rating 4 $\rightarrow$ 10%
- Rating 3 $\rightarrow$ 5%
- Rating 1-2 $\rightarrow$ no bonus

Attendance Condition:

- If absences are more than 5, reduce the calculated bonus by 50%.
- If absences are more than 10, no bonus is given.

Calculate:

$\text{Bonus} = \text{Salary} \times \text{Bonus Rate}$
$\text{Total Pay} = \text{Salary} + \text{Bonus}$

Problem 53: Restaurant Bill With Senior Discount

Objective: Calculate a restaurant bill with service charge and senior discount.

Inputs:

- Food bill
- Senior citizen status (y/n)

Conditions:

- Food bill ₱1,000 or above → add 10% service charge.
- Food bill below ₱1,000 → no service charge.
- Senior citizen → apply 20% discount after adding the service charge.

Calculate:

$\text{Subtotal} = \text{Food Bill} + \text{Service Charge}$
$\text{Final Bill} = \text{Subtotal} - \text{Discount}$

Problem 54: Online Shopping Checkout

Objective: Calculate a customer's final checkout amount.

Inputs:

- Purchase amount
- Membership status (y/n)
- Coupon status (y/n)

Conditions:

- Purchase $\geq$ ₱2,000 → 10% store discount
- Member → additional 5% discount
- Valid coupon → subtract ₱100
- Orders below ₱1,500 after discounts → add ₱120 shipping
- Orders ₱1,500 or above → free shipping

The final amount cannot be negative.

Problem 55: Loan Approval System

Objective: Determine whether a loan applicant is qualified.

Inputs:

- Monthly salary
- Years employed
- Existing monthly debt

Calculate:

```
Debt Ratio = Existing Debt / Monthly Salary × 100
```

Conditions:

- Salary $\geq$ ₱25,000
- Years employed $\geq$ 2
- Debt ratio $\leq$ 40%

If all three are true:

```
Loan Approved
```

Otherwise:

```
Loan Denied
```

Also display which requirement was not met.

Problem 56: College Admission Checker

Objective: Determine whether a student qualifies for admission.

Inputs:

- Entrance exam score
- High school average
- Interview score

Conditions:

- Entrance score $\geq$ 80
- High school average $\geq$ 85
- Interview score $\geq$ 75

If all conditions are met → **Accepted**

If entrance score $\geq$ 75 and average $\geq$ 80 but one requirement is missed → **Waitlisted**

Otherwise → **Rejected**

Problem 57: Electricity Bill With Senior and Late Fee

Objective: Calculate an electricity bill with discounts and penalties.

Inputs:

- kWh consumed
- Senior status (y/n)
- Payment status (on-time/late)

Rates:

- 1-100 kWh → ₱9/kWh
- 101-200 kWh → ₱11/kWh
- Above 200 kWh → ₱14/kWh

Additional Conditions:

- Senior citizen → 10% discount
- Late payment → 5% surcharge after the discount

Display:

- Basic bill
- Senior discount
- Late fee
- Final bill

Problem 58: Employee Overtime With Weekend Rate

Objective: Calculate overtime pay depending on the type of day.

Inputs:

- Basic hourly rate
- Regular hours
- Overtime hours
- Is it a weekend? (y/n)

Conditions:

- Regular hours → normal hourly rate
- Weekday overtime → $1.5 \times$ hourly rate
- Weekend overtime → $2 \times$ hourly rate

Calculate:

Regular Pay + Overtime Pay = Total Pay

Problem 59: Delivery Charge With Priority Service

Objective: Calculate a delivery fee using distance, weight, and delivery priority.

Inputs:

- Distance in km
- Weight in kg
- Priority delivery (y/n)

Distance Charges:

- Up to 5 km → ₱50
- 6–15 km → ₱100
- Above 15 km → ₱150

Weight Charges:

- Up to 3 kg → ₱0
- 3.01–10 kg → ₱75
- Above 10 kg → ₱150

Priority Delivery:

- *y* → add 25% to the total delivery charge

Problem 60: Hotel Booking Calculator

Objective: Calculate the total hotel charge.

Inputs:

- Room type
- Number of nights
- Weekend stay (*y/n*)

Room Rates:

- Standard → ₱1,500/night
- Deluxe → ₱2,500/night
- Suite → ₱4,000/night

Conditions:

- 5–9 nights → 5% discount
- 10 nights or more → 10% discount
- Weekend stay → add ₱300 per night

Display the original charge, weekend surcharge, discount, and final bill.

Problem 61: ATM Transaction Validator

Objective: Validate and process an ATM withdrawal.

Inputs:

- Account balance
- Withdrawal amount

Conditions:

- Withdrawal must be greater than ₱0.
- Withdrawal must be divisible by ₱100.
- Withdrawal cannot exceed ₱20,000 per transaction.
- Balance after withdrawal must remain at least ₱500.

If all conditions are satisfied, process the withdrawal.

Otherwise, display the appropriate error.

Problem 62: Scholarship and Allowance Calculator

Objective: Determine scholarship level and monthly allowance.

Inputs:

- General average
- Family monthly income

- Conduct grade

Conditions:

- Average ≥ 95 , income $\leq \text{P}20,000$, conduct $\geq 90 \rightarrow$ Full Scholarship + $\text{P}3,000$ allowance
- Average ≥ 90 , income $\leq \text{P}30,000$, conduct $\geq 85 \rightarrow$ 75% Scholarship + $\text{P}2,000$ allowance
- Average ≥ 85 , income $\leq \text{P}40,000$, conduct $\geq 80 \rightarrow$ 50% Scholarship + $\text{P}1,000$ allowance
- Otherwise $\rightarrow$ No Scholarship

Problem 63: Phone Plan With Data and Calls

Objective: Calculate a phone bill using both data and call usage.

Inputs:

- GB used
- Call minutes used

Base Plan: $\text{P}500$

Data Charge:

- Up to 10 GB $\rightarrow \text{P}0$
- 11-20 GB $\rightarrow \text{P}200$
- Above 20 GB $\rightarrow \text{P}500$

Call Charge:

- Up to 100 minutes $\rightarrow \text{P}0$
- Above 100 minutes $\rightarrow \text{P}2$ per excess minute

Calculate:

Final Bill = Base Plan + Data Charge + Call Charge

Problem 64: Employee Tax and Insurance Calculator

Objective: Calculate deductions and net salary.

Input:

Monthly salary

Tax:

- Up to $\text{P}20,000 \rightarrow 0\%$
- $\text{P}20,001$ - $\text{P}40,000 \rightarrow 5\%$
- $\text{P}40,001$ - $\text{P}70,000 \rightarrow 10\%$
- Above $\text{P}70,000 \rightarrow 15\%$

Insurance:

- Salary below $\text{P}30,000 \rightarrow \text{P}500$
- Salary $\text{P}30,000$ or above $\rightarrow \text{P}1,000$

Calculate:

Net Salary = Salary - Tax - Insurance

Problem 65: Grocery Checkout With Discounts

Objective: Calculate a grocery bill using customer type and purchase total.

Inputs:

- Grocery total
- Customer type: `regular`, `senior`, or `member`

Conditions:

- Regular → no customer discount
- Member → 5% discount
- Senior → 10% discount

After customer discount:

- If total is still ₱3,000 or more → additional ₱200 discount

Display all discounts separately.

Problem 66: Traffic Violation Fine

Objective: Calculate a driver's total fine.

Inputs:

- Violation type
- Number of previous violations

Base Fines:

- Speeding → ₱1,000
- Illegal Parking → ₱500
- No Seatbelt → ₱750

Repeat-Offender Conditions:

- 0 previous violations → regular fine
- 1-2 previous violations → add 25%
- 3 or more → add 50%

Display the base fine, penalty, and total fine.

Problem 67: Gym Membership Calculator

Objective: Calculate gym membership cost.

Inputs:

- Membership type
- Number of months
- Student status (`y/n`)

Monthly Rates:

- Basic → ₱800
- Premium → ₱1,200
- VIP → ₱2,000

Conditions:

- 6 months or more → 5% discount
- 12 months or more → 10% discount
- Student → additional 5% discount

Calculate the final membership fee.

Problem 68: Fuel Purchase Discount

Objective: Calculate the total fuel cost.

Inputs:

- Liters purchased
- Price per liter
- Loyalty member (y/n)

Calculate:

Basic Cost = Liters × Price per Liter

Discounts:

- 20–49 liters → 2%
- 50 liters or more → 5%
- Loyalty member → additional 2%

Display original price, total discount, and final cost.

Problem 69: Cargo Shipping Calculator

Objective: Calculate a cargo shipping fee.

Inputs:

- Weight
- Destination type: `local`, `provincial`, or `international`
- Express shipping (y/n)

Rate per kg:

- Local → ₱20
- Provincial → ₱40
- International → ₱100

Conditions:

- Weight above 20 kg → 10% heavy-cargo surcharge
- Express shipping → additional 25%

Calculate the final shipping cost.

Problem 70: Student Tuition Calculator

Objective: Calculate tuition after scholarship and miscellaneous fees.

Inputs:

- Number of units
- Cost per unit
- Scholarship percentage
- Laboratory subject (y/n)

Calculate:

```
Tuition = Units × Cost per Unit
```

Conditions:

- Laboratory subject → add ₱2,000 lab fee
- Apply scholarship only to the tuition, not to the laboratory fee.

Formula:

```
Final Tuition = Discounted Tuition + Lab Fee
```

Validate that the scholarship percentage is between 0 and 100.

Problem 71: Car Rental Calculator

Objective: Calculate the total car-rental charge.

Inputs:

- Car type
- Number of rental days
- Distance traveled

Daily Rates:

- Economy → ₱1,500
- Sedan → ₱2,500
- SUV → ₱4,000

Mileage Condition:

- First 100 km per rental → free
- Excess distance → ₱15/km

Discount:

- 7 days or more → 10% discount on rental fee only

Calculate rental fee, mileage charge, discount, and final cost.

Problem 72: Employee Promotion Eligibility

Objective: Determine whether an employee qualifies for promotion.

Inputs:

- Years of service
- Performance rating
- Number of absences
- Number of disciplinary warnings

Conditions:

Employee qualifies if:

- Years of service ≥ 3
- Performance rating ≥ 4
- Absences ≤ 5
- Disciplinary warnings = 0

If all conditions are met:

Eligible for Promotion

Otherwise:

Not Eligible

Display the failed qualification requirements.

Problem 73: Hospital Billing Calculator

Objective: Calculate a simplified hospital bill.

Inputs:

- Number of hospital days
- Room type
- Senior status (y/n)

Room Rates Per Day:

- Ward → ₱1,000
- Private → ₱2,500
- Suite → ₱5,000

Additional Charges:

- Medical fee = ₱3,000
- Hospital stay over 7 days → add ₱2,000 extended-stay fee

Senior Discount:

- Senior citizen → 15% discount on the room charge only.

Calculate and display each charge and the final amount.

Problem 74: Food Delivery Order Calculator

Objective: Calculate a customer's final food-delivery bill.

Inputs:

- Food order total
- Distance
- Premium member (y/n)
- Promo code (y/n)

Delivery Fee:

- Up to 3 km → ₱50
- 4-10 km → ₱100
- Above 10 km → ₱150

Conditions:

- Food order $\geq$ ₱1,500 → delivery is free
- Premium member → 5% discount on food
- Valid promo → additional ₱100 discount

The final total cannot fall below ₱0.

Problem 75: Complete Employee Payroll System

Objective: Create a payroll program combining basic salary, overtime, bonus, tax, and attendance deductions.

Inputs:

- Basic salary
- Overtime hours
- Performance rating
- Number of absences

Overtime:

- 0-10 hours → ₱100/hour
- 11-20 hours → ₱125/hour
- Above 20 hours → ₱150/hour

Performance Bonus:

- Rating 5 → 10% of basic salary
- Rating 4 → 5%
- Rating 1-3 → no bonus

Attendance Deduction:

- 0-2 absences → ₱0
- 3-5 absences → ₱500
- More than 5 absences → ₱1,500

Calculate:

Gross Pay = Basic Salary + Overtime Pay + Bonus

Tax:

- Gross pay $\leq$ ₱20,000 $\rightarrow$ 0%
- ₱20,001–₱40,000 $\rightarrow$ 5%
- ₱40,001–₱60,000 $\rightarrow$ 10%
- Above ₱60,000 $\rightarrow$ 15%

Then:

$$\text{Tax} = \text{Gross Pay} \times \text{Tax Rate}$$

$$\text{Net Pay} = \text{Gross Pay} - \text{Tax} - \text{Attendance Deduction}$$

Test Cases:

Salary	OT	Rating	Absences	Main Conditions
₱18,000	5	3	1	OT, no bonus, no absence deduction
₱25,000	10	4	3	OT + 5% bonus + ₱500 deduction
₱35,000	15	5	0	OT + 10% bonus
₱50,000	20	4	6	OT + bonus + ₱1,500 deduction
₱70,000	25	5	2	Highest tax bracket

This set now forces students to combine calculations rather than just choose one branch. For Problems 76–100, the strongest progression would be mini real-world systems with nested decisions, input validation, multiple interacting conditions, logical operators, and more complex billing/payroll/order calculations.

SECTION IV

Problems 76-100: Multi-Stage Systems

Realistic, multi-stage calculations that combine several conditional rules.

Problem 76: Supermarket Checkout System

Objective: Calculate a customer's final supermarket bill using purchase amount, membership, and payment method.

Inputs:

- Total purchase amount
- Member? (y/n)
- Payment method (cash/card)

Discount:

- Below ₱1,000 → 0%
- ₱1,000–₱2,999 → 5%
- ₱3,000–₱4,999 → 10%
- ₱5,000+ → 15%

Additional Conditions:

- Member → additional 5% discount on the amount after the first discount.
- Card payment → add 2% processing fee after discounts.
- Cash → no processing fee.

Display: Original amount, discounts, processing fee, and final bill.

Problem 77: Advanced Electricity Billing

Objective: Calculate an electricity bill with consumption rates, senior discount, and late-payment penalty.

Inputs:

- kWh consumed
- Senior citizen? (y/n)
- Payment late? (y/n)

Rates:

- 0 or below → ₱0
- 1–100 → ₱9/kWh
- 101–200 → ₱11/kWh
- 201–300 → ₱13/kWh
- Above 300 → ₱16/kWh

Conditions:

- Senior citizen → 10% discount

- Late payment → 8% surcharge after discount
- Final bill cannot be negative.

Test Cases:

kWh	Senior	Late	Basic Bill
80	n	n	₱720
150	n	n	₱1,650
250	y	n	₱3,250
350	n	y	₱5,600

Problem 78: Advanced Employee Payroll

Objective: Calculate an employee's complete salary.

Inputs:

- Hours worked
- Hourly rate
- Performance rating
- Absences

Pay Rules:

- First 40 hours → normal rate
- Hours above 40 → 1.5× rate

Bonus:

- Rating 5 → 10% of regular + overtime pay
- Rating 4 → 5%
- Below 4 → no bonus

Absence Deduction:

- 0-2 → none
- 3-5 → ₱500
- Above 5 → ₱1,500

Calculate:

Gross Pay = Regular Pay + Overtime Pay + Bonus

Net Pay = Gross Pay - Absence Deduction

Problem 79: University Admission System

Objective: Determine an applicant's admission status.

Inputs:

- Entrance exam score
- High school average
- Interview score

- Has disciplinary record? (y/n)

Conditions:

- Exam ≥ 90 AND average ≥ 90 AND interview ≥ 85 AND no disciplinary record → [Priority Admission](#)
- Exam ≥ 80 AND average ≥ 85 AND interview ≥ 75 AND no disciplinary record → [Regular Admission](#)
- Exam ≥ 75 AND average ≥ 80 → [Waitlisted](#)
- Otherwise → [Rejected](#)

A disciplinary record prevents Priority and Regular Admission.

Problem 80: Online Store Checkout

Objective: Calculate the final amount for an online order.

Inputs:

- Order amount
- Member? (y/n)
- Promo code? (y/n)
- Delivery distance

Conditions:

- Member → 10% discount
- Promo code → additional ₱150 discount

Shipping:

- Order after discounts $\geq$ ₱2,000 → free
- Otherwise:
 - Up to 5 km → ₱80
 - 6–15 km → ₱150
 - Above 15 km → ₱250

Final amount cannot be below zero.

Problem 81: Bank Loan Evaluation

Objective: Determine whether a customer qualifies for a loan and calculate the maximum loan amount.

Inputs:

- Monthly income
- Monthly debt
- Years employed
- Credit score

Calculate:

$$\text{Debt Ratio} = \text{Monthly Debt} / \text{Monthly Income} \times 100$$

Conditions:

- Credit score ≥ 750 , debt ratio $\leq 30\%$, employment ≥ 3 years → [Excellent](#)

- Credit score ≥ 650 , debt ratio $\leq 40\%$, employment ≥ 2 years $\rightarrow$ **Qualified**
- Credit score ≥ 600 and debt ratio $\leq 45\%$ $\rightarrow$ **Conditional Approval**
- Otherwise $\rightarrow$ **Rejected**

Maximum Loan:

- Excellent $\rightarrow$ income $\times 20$
- Qualified $\rightarrow$ income $\times 12$
- Conditional $\rightarrow$ income $\times 6$

Validate that monthly income is greater than zero.

Problem 82: Advanced Hotel Reservation

Objective: Calculate a hotel reservation bill.

Inputs:

- Room type
- Number of nights
- Weekend nights
- Member? (y/n)

Rates:

- Standard $\rightarrow$ ₱2,000/night
- Deluxe $\rightarrow$ ₱3,500/night
- Suite $\rightarrow$ ₱5,500/night

Conditions:

- Weekend night $\rightarrow$ additional ₱500 per weekend night
- Stay 5–9 nights $\rightarrow$ 5% room discount
- Stay 10+ nights $\rightarrow$ 10% room discount
- Member $\rightarrow$ additional 5% discount

Validate that weekend nights cannot exceed total nights.

Problem 83: Airline Baggage Fee Calculator

Objective: Calculate baggage fees based on passenger and baggage information.

Inputs:

- Ticket class (economy/business)
- Baggage weight
- Senior citizen? (y/n)

Free Baggage:

- Economy $\rightarrow$ first 20 kg free
- Business $\rightarrow$ first 30 kg free

Excess Fee:

- ₱200 per excess kilogram

Conditions:

- Senior → 20% discount on excess baggage fee
- No excess weight → ₱0 fee

Example: Economy, 25 kg, non-senior → $5 \times \text{₱}200 = \text{₱}1,000$.

Problem 84: Advanced Taxi Fare

Objective: Calculate a taxi fare based on distance, traffic, and time.

Inputs:

- Distance traveled
- Traffic level (*normal*/*heavy*)
- Night trip? (*y*/*n*)

Base Fare: ₱50 for first 3 km.

Additional Distance: ₱15/km beyond 3 km.

Conditions:

- Heavy traffic → add 10%
- Night trip → add 20%
- If both apply, calculate both surcharges from the fare before surcharges.

Display base fare, distance charge, surcharges, and total.

Problem 85: Scholarship Ranking System

Objective: Determine a student's scholarship category.

Inputs:

- General average
- Entrance exam score
- Family income
- Conduct grade

Conditions:

- Average ≥ 95 AND exam ≥ 90 AND income $\leq \text{₱}25,000$ AND conduct ≥ 90 → *100% Scholarship*
- Average ≥ 90 AND exam ≥ 85 AND income $\leq \text{₱}40,000$ AND conduct ≥ 85 → *75% Scholarship*
- Average ≥ 85 AND exam ≥ 80 AND income $\leq \text{₱}60,000$ AND conduct ≥ 80 → *50% Scholarship*
- Otherwise → *No Scholarship*

Problem 86: Restaurant Ordering System

Objective: Calculate a restaurant customer's complete bill.

Inputs:

- Food subtotal
- Dine-in or takeout
- Senior? (y/n)
- Number of diners

Conditions:

- Dine-in → 10% service charge
- Takeout → ₱50 packaging fee
- Senior → 20% discount on food subtotal only
- 8 or more diners → additional 5% group charge on food subtotal

Calculate all charges separately and display the final bill.

Problem 87: Insurance Premium Calculator

Objective: Calculate a simplified vehicle insurance premium.

Inputs:

- Vehicle value
- Driver age
- Number of previous accidents

Base Premium:

Vehicle Value × 3%

Age Adjustment:

- Under 25 → add 25%
- 25-59 → no age surcharge
- 60+ → add 15%

Accident Adjustment:

- 0 accidents → no surcharge
- 1 accident → add 10%
- 2+ accidents → add 25%

All percentage surcharges are calculated from the base premium.

Problem 88: Internet Café Billing

Objective: Calculate a customer's computer-use bill.

Inputs:

- Number of hours
- Student? (y/n)
- Printing pages

Computer Rate:

- First 2 hours → ₱40/hour
- Next 3 hours → ₱30/hour
- Hours above 5 → ₱25/hour

Printing: ₱5/page

Discount:

- Student → 10% discount on computer charge only

Calculate computer charge, printing charge, discount, and final bill.

Problem 89: Employee Promotion Score

Objective: Determine promotion eligibility using a calculated score.

Inputs:

- Performance rating (1-5)
- Years of service
- Attendance percentage
- Training courses completed

Calculate:

```
Score = (Performance × 10) + (Years × 2) + Training
```

Conditions:

- Score ≥ 70 AND attendance ≥ 95% → **Highly Recommended**
- Score ≥ 55 AND attendance ≥ 90% → **Recommended**
- Score ≥ 45 AND attendance ≥ 85% → **For Review**
- Otherwise → **Not Recommended**

Problem 90: Water Billing With Conservation Discount

Objective: Calculate a household's water bill and reward low consumption.

Inputs:

- Water consumption
- Previous month's consumption

Current Rates:

- Up to 20 m³ → ₱25/m³
- 21-40 m³ → ₱35/m³
- Above 40 m³ → ₱50/m³

Calculate percentage reduction from the previous month.

Conditions:

- Consumption reduced by at least 20% → 10% conservation discount
- Reduced by 10-19.99% → 5% discount

- Otherwise → no discount

Validate that previous consumption is greater than zero.

Problem 91: Credit Card Payment Calculator

Objective: Determine charges for a credit-card payment.

Inputs:

- Current balance
- Payment amount
- Payment late? (y/n)

Conditions:

- Payment $\geq$ balance → no interest
- Payment $\geq$ 50% of balance → 2% interest on remaining balance
- Payment below 50% → 5% interest on remaining balance
- Late payment → additional ₱500 penalty

Calculate:

Remaining Balance = Balance - Payment
New Balance = Remaining Balance + Interest + Late Fee

Payment cannot exceed the balance.

Problem 92: Advanced Parking System

Objective: Calculate parking fees based on vehicle type and duration.

Inputs:

- Vehicle type (motorcycle/car/van)
- Hours parked
- Lost ticket? (y/n)

First 3 Hours:

- Motorcycle → ₱30
- Car → ₱50
- Van → ₱80

Additional Hour:

- Motorcycle → ₱10
- Car → ₱20
- Van → ₱30

Condition:

- Lost ticket → add ₱500 penalty.

Example: Car parked 5 hours → ₱50 + (2 × ₱20) = ₱90.

Problem 93: E-Commerce Seller Fee Calculator

Objective: Calculate a seller's net earnings.

Inputs:

- Product selling price
- Product category
- Shipping subsidy? (y/n)

Platform Fees:

- Electronics → 8%
- Clothing → 10%
- Food → 5%
- Other → 12%

Additional Conditions:

- Payment-processing fee = 2% of selling price
- Shipping subsidy → seller pays an additional ₱50

Calculate:

Net Earnings = Selling Price - Platform Fee - Processing Fee - Shipping Cost

Problem 94: Advanced Tuition Assessment

Objective: Calculate a student's total school fees.

Inputs:

- Number of units
- Rate per unit
- Laboratory units
- Scholarship type

Base Tuition:

Units × Rate per Unit

Laboratory Fee:

Laboratory Units × ₱500

Scholarships:

- **full** → 100% tuition discount
- **half** → 50% tuition discount
- **quarter** → 25% tuition discount
- **none** → 0%

Scholarship applies only to base tuition.

Add:

- Registration fee = ₱1,000
- Library fee = ₱500

Calculate the final assessment.

Problem 95: Food Delivery Rider Earnings

Objective: Calculate a rider's daily earnings.

Inputs:

- Number of deliveries
- Total kilometers traveled
- Customer rating

Earnings:

- ₱50 per delivery
- ₱5 per kilometer

Bonuses:

- 20+ deliveries → ₱500 bonus
- 10-19 deliveries → ₱200 bonus

Rating Bonus:

- Rating ≥ 4.8 → additional ₱300
- Rating 4.5-4.79 → additional ₱150
- Below 4.5 → no rating bonus

Calculate total daily earnings.

Problem 96: Advanced Sales Commission

Objective: Calculate an employee's total monthly earnings.

Inputs:

- Basic salary
- Monthly sales
- Customer rating

Commission:

- Sales below ₱50,000 → 2%
- ₱50,000-₱99,999 → 5%
- ₱100,000-₱199,999 → 8%
- ₱200,000+ → 12%

Customer Rating Bonus:

- Rating ≥ 4.8 → ₱2,000
- Rating ≥ 4.5 → ₱1,000
- Otherwise → ₱0

Calculate:

Total Earnings = Basic Salary + Commission + Rating Bonus

Problem 97: Vehicle Rental With Fuel Charge

Objective: Calculate the complete cost of renting a vehicle.

Inputs:

- Vehicle type
- Number of days
- Kilometers traveled
- Fuel tank returned full? (y/n)

Daily Rates:

- Motorcycle → ₱800
- Sedan → ₱2,000
- SUV → ₱3,500

Mileage:

- First 200 km → free
- Excess → ₱10/km

Conditions:

- 7+ days → 10% discount on rental charge
- Fuel tank not full → add ₱1,500 fuel charge

Display all charges separately.

Problem 98: Employee Leave Approval System

Objective: Determine whether an employee's leave request should be approved.

Inputs:

- Available leave credits
- Requested leave days
- Number of absences this year
- Is request an emergency? (y/n)

Conditions:

- Requested days $\leq$ leave credits AND absences ≤ 5 → **Approved**
- Emergency AND requested days $\leq$ leave credits → **Approved – Emergency**
- Requested days $>$ leave credits → **Denied – Insufficient Leave Credits**
- Otherwise → **Denied – Attendance Record**

If approved:

Remaining Credits = Available Credits - Requested Days

Problem 99: Complete Utility Bill Calculator

Objective: Calculate a household's combined electricity, water, and internet bill.

Inputs:

- Electricity consumption in kWh
- Water consumption in m³
- Internet data usage in GB
- Senior citizen? (y/n)

Electricity:

- Up to 100 kWh → ₱10/kWh
- 101-200 → ₱12/kWh
- Above 200 → ₱15/kWh

Water:

- Up to 20 m³ → ₱25/m³
- 21-40 → ₱30/m³
- Above 40 → ₱40/m³

Internet:

- Up to 100 GB → ₱999
- 101-300 GB → ₱1,299
- Above 300 GB → ₱1,799

Calculate:

```
Subtotal = Electricity + Water + Internet
```

Condition:

- Senior citizen → 10% discount on electricity and water only.
- Internet receives no senior discount.

Display:

- Electricity bill
- Water bill
- Internet bill
- Senior discount
- Final utility bill

Problem 100: Complete Point-of-Sale System

Objective: Create a Python program that simulates a small store checkout system using multiple calculations and conditional statements.

Inputs:

- Product price
- Quantity
- Customer type (regular/member/senior)
- Payment method (cash/card)
- Promo code? (y/n)

Calculate:

$$\text{Subtotal} = \text{Product Price} \times \text{Quantity}$$

Step 1: Quantity Discount

- Subtotal below ₱1,000 → 0%
- ₱1,000–₱2,999 → 5%
- ₱3,000–₱4,999 → 10%
- ₱5,000 and above → 15%

Calculate:

$$\text{Amount After Quantity Discount} = \text{Subtotal} - \text{Quantity Discount}$$

Step 2: Customer Discount

- Regular → 0%
- Member → additional 5%
- Senior → additional 10%

Apply this percentage to the amount remaining after the quantity discount.

Step 3: Promo Code

If the customer has a valid promo code:

- Subtract another ₱100.

Otherwise:

- No promo discount.

Step 4: Payment Fee

If payment method is:

- **cash** → no processing fee
- **card** → add 2% processing fee

Calculate the processing fee after all discounts.

Step 5: Final Bill

Calculate:

$$\text{Final Bill} = \text{Discounted Amount} + \text{Processing Fee}$$

The final bill cannot be below ₱0.

Required Output

Display:

- Original subtotal
- Quantity discount
- Customer discount
- Promo discount
- Processing fee
- Final bill

Test Cases

Price	Qty	Customer	Payment	Promo
₱100	5	Regular	Cash	n
₱500	2	Member	Cash	n
₱1,000	3	Member	Card	y
₱1,000	5	Senior	Cash	y
₱2,000	5	Senior	Card	y

Example for Test Case 3:

Product price = ₱1,000

Quantity = 3

Subtotal = ₱1,000 × 3 = ₱3,000

10% quantity discount:

₱3,000 × 10% = ₱300

Amount remaining:

₱3,000 - ₱300 = ₱2,700

5% member discount:

₱2,700 × 5% = ₱135

After member discount:

₱2,700 - ₱135 = ₱2,565

Promo discount:

₱2,565 - ₱100 = ₱2,465

2% card processing fee:

₱2,465 × 2% = ₱49.30

Final Bill = ₱2,514.30